

AI for Safer Roads Innovation Challenge 2026

Speed Limit Misalignment Model

This methodology document follows the exact flow of the notebook and organizes it as a reviewer-friendly report. Each notebook section is retained, with the corresponding output table, chart, or map interpretation placed beside the method that produced it. Large notebook outputs such as the full scored CSV, full interactive HTML map, and top 300 road list are described in the output guide and exported as separate files, while key excerpts are shown here for readability.

Purpose of this notebook: build a clear, reproducible, policy-ready model that helps road agencies identify where **posted speed limits may be misaligned with real-world road conditions**.

Business problem

Road agencies need to know **which road segments should be reviewed first** because their posted speed limits may be too high for the surrounding land use, road function, operating speeds, and likely vulnerable-road-user exposure.

Solution

We combine Safe System speed guidance, road function, urban/rural land use, operating speeds, traffic exposure, and an interpretable AI/ML peer model to create a **Speed Safety Score, Priority Score, priority band, and geospatial outputs** that governments can use for review and intervention planning.

How this methodology aligns with the challenge requirements

The ADB challenge asks for a solution that can assess Safe System alignment, identify misaligned speed limits, produce policy-ready spatial outputs, and be scalable across countries. This notebook is structured around those requirements.

Challenge requirement	How this notebook answers it
Safe Speed Assessment	Compares posted speed limits against Safe System reference speeds based on road class and land use.
Risk Identification	Flags roads where posted limits and operating speeds exceed the reference speed, especially in higher vulnerable-road-user environments.
Policy-Ready Outputs	Produces `PriorityScore`, `SpeedSafetyScore`, `PriorityBand`, recommendations, CSV, GeoJSON, and an interactive HTML map.
Scalable and replicable across countries	Uses the same standardized field schema and scoring workflow for Maharashtra and Thailand.
Transparent and interpretable	Shows the contribution of each component: posted-limit mismatch, operating-speed context, VRU proxy, traffic exposure, and peer-model gap.
Methodologically robust	Includes data validation, filtering audit, model diagnostics, sensitivity testing, and reproducibility notes.

Source references used in the narrative and methodology:

- ADB AI for Safer Roads Innovation Challenge official page and criteria.
- AI for Safer Roads 2026 Data User Guide v1.0.
- Safe System / speed-management guidance from WHO, World Bank / GRSF, ITF/OECD, and iRAP.

Data sources and modelling philosophy

Data provided

The model uses two GeoJSON files:

1. ADB_Innovation_Maharashtra.geojson
2. ADB_Innovation_Thailand.geojson

The Data User Guide explains that the road network comes from Overture/OpenStreetMap-based road sections and is enriched with TomTom-derived speed, probe, and traffic-exposure indicators. It also explains that the files are backend data for an online dashboard and that **many fields are not required**. The most important fields are generally those relating to **sample size and vehicle speeds**.

Modelling philosophy

We use a **hybrid approach**:

1. **Rules grounded in Safe System thinking** for transparent, policy-aligned screening.
2. **Geospatial analytics** to preserve road-segment location and enable map-based review.
3. **AI/ML peer model** as a supporting signal to identify speed limits that look high compared with similar road contexts.

The model intentionally keeps the AI component small in the final score because the dataset does not include all engineering factors that usually determine a legally appropriate speed limit, such as median barriers, pedestrian crossings, school zones, markets, lane width, roadside activity, or access control.

Reproducibility checklist

To rerun the notebook from scratch:

1. Place the two GeoJSON files in the same folder as this notebook, or set the `SAFER_ROADS_DATA_DIR` environment variable to the folder containing the files.
2. Run all cells from top to bottom.
3. Outputs will be written to a folder called `safer_roads_submission_outputs`.

Expected Python packages

- pandas
- numpy
- geopandas
- shapely
- matplotlib
- scikit-learn
- folium

The workflow uses a fixed random seed so the AI peer model and diagnostics are reproducible.

Notebook section label: 2.1 Setup

Notebook output

```
Search roots:  
- /content
```

```
Outputs will be saved to: /content/safer_roads_submission_outputs
```

Notebook section label: 2.2 Locate input files / Find the newest file matching any of the patterns across search roots.

Notebook output

```
Maharashtra file: /content/ADB_Innovation_Maharashtra.geojson  
Thailand file: /content/ADB_Innovation_Thailand.geojson
```

Load the GeoJSON files

We load both datasets as GeoDataFrames so the geometry is preserved for mapping and GeoJSON export.

Notebook section label: 3. Load data

Raw data summary

Country	Raw rows	Raw columns
Maharashtra	14082	29
Thailand	55884	28
Total	69966	

Interpretation: The raw input contains 69,966 road rows across Maharashtra and Thailand. Thailand has the larger raw file, while both countries preserve geometry through GeoDataFrames so they can later be exported as map-ready layers.

Notebook section label: Quick column audit: which fields appear in each file?

Column audit by country

Column	In Maharashtra	In Thailand
AnalysisStatus	True	True
Country	True	True
DISSOLVE ID	True	False
ExcludeFromSpeedSPI	True	False
F85thPercentileSpeed	True	True
ForAnalysis	False	True
InvPercentile	False	True
LandUse	True	True
MedianSpeed	True	True
NO_OF_Result_Segments	False	True
NumberOverLimit	True	True
OBJECTID	True	True
OvertureID	False	True
Pass	True	False
PercentOverLimit	True	True
Percent	True	True
Percentile	True	True
PercentileBand	True	True
ProvinceID	False	True
RankedPercentile	True	True
RoadClass	True	True
RoadLength	True	True
SampleSizeTotal	False	True
SampleSize_avg	True	True
Sample_Size_Total	True	False
Shape_Length	True	True
SpeedLimit	True	True
SpeedLimitFloor	True	True

StreetImageLink	True	True
UrbanPC	True	False
WeightedSample	True	True
class	True	False
english_ro	False	True
geometry	True	True
names_primary	True	False
subtype	True	False

Field standardization: what was included, standardized, and excluded

The two files are broadly similar, but some columns use different names for the same concept. We standardize those columns so one workflow can run across both countries.

Key standardization decisions

- `names_primary` and `english_ro` both represent the road name, so both become `RoadName`.
- `Sample_Size_Total` and `SampleSizeTotal` both represent total sample size, so both become `SampleSizeTotal` and later `SampleSizeUnified`.
- `DISSOLVE_ID` and `OvertureID` are source-specific identifiers, so we keep them as `SourceRoadID` but create our own stable `SegmentID` from `Country` + `OBJECTID`.
- The model uses `Shape_Length`, not `RoadLength`, because the Data User Guide says to ignore `RoadLength` and use shape length.

Why not include every field?

We only keep fields that help answer the challenge problem: is the posted limit appropriate for the road environment? Fields marked "ignore," dashboard-only fields, country-only administrative fields, or duplicate raw columns are excluded from scoring.

Notebook section label: 4.1 Included / standardized fields dictionary

Included and standardized fields dictionary

Standardized field	Original field/s	Plain-English meaning	Why included
Country	Created	Source country / area	Needed for country-level summaries and country-specific peer models.
SegmentID	Created from Country + OBJECTID	Stable model segment ID	Makes IDs unique after combining countries.
SourceRoadID	DISSOLVE_ID / OvertureID	Original source road ID	Kept for traceability to source data.
OBJECTID	OBJECTID	GIS-created unique ID	Useful for tracing a result back to the original file.
RoadName	names_primary / english_ro	Road name	Useful for business users, maps, and top-priority lists.
RoadClass	RoadClass	Motorway / trunk / primary / secondary	Core model field. Reference speed depends on road function.
LandUse	LandUse	Urban / rural	Core model field. Reference speed and VRU exposure differ by land use.
SpeedLimit	SpeedLimit	Posted speed limit	Main field being assessed for possible misalignment.
MedianSpeed	MedianSpeed	50th percentile operating speed	Operating-speed context; helps understand real-world speeds.
F85thPercentileSpeed	F85thPercentileSpeed	85th percentile operating speed	Key operating-speed indicator; used to check whether the environment is oper...
PercentOverLimit	PercentOverLimit	Estimated share over posted limit	Used as context, not as the main scoring rule, to avoid becoming a driver-en...
WeightedSample	WeightedSample	Traffic exposure weighted by road length	Prioritizes review where more travel occurs.
SampleSizeUnified	Sample_Size_Total / SampleSizeTotal / SampleSize_avg	Unified traffic sample size	Confidence/context field for model diagnostics and peer model features.
RankedPercentile	RankedPercentile	Road's share/position in traffic exposure	Converted into TrafficExposureScore for prioritization.
StreetImageLink	StreetImageLink	Lat/lon for street-level validation	Useful for field validation and future image review, but not scored directly...
Shape_Length	Shape_Length	Geometric segment length	Used for length summaries and geospatial outputs.
geometry	geometry	Road segment geometry	Required for GeoJSON export and interactive map.

AnalysisStatus	AnalysisStatus	Internal validation flag	Used to filter to records marked valid.
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Notebook section label: 4.2 Excluded fields dictionary

Excluded fields dictionary

Excluded / not directly scored field	Reason
RoadLength	Data guide says ignore and use Shape Length instead.
Percent	Data guide says ignore.
ForAnalysis	Data guide says ignore: AnalysisStatus is used for validation filtering inst...
ProvinceID	Thailand-only administrative field; guide says not useful for this analysis.
SpeedLimitFloor	Data guide says ignore.
NO OF Result Segments	Data guide says ignore.
InvPercentile	Dashboard filtering field only; not needed for model scoring.
PercentileBand / percentileBand	Dashboard category; numeric RankedPercentile is better for scoring.
Percentile	Raw percentile retained only for audit if needed; RankedPercentile is cleare...
NumberOverLimit	Not used in final score to avoid making the model a speeding-enforcement mod...
DISSOLVE ID	Maharashtra-only source ID; standardized into SourceRoadID.
OvertureID	Thailand source ID; standardized into SourceRoadID.
names_primary	Maharashtra road-name field; standardized into RoadName.
english_ro	Thailand road-name field; standardized into RoadName.
Sample_Size_Total	Maharashtra sample-size field; standardized into SampleSizeUnified.
SampleSizeTotal	Thailand sample-size field; standardized into SampleSizeUnified.
class	Raw Overture/source classification; RoadClass is the cleaner modelling field.
subtype	Road subtype label, not needed for the score with available Phase 1 fields.
UrbanPC	Maharashtra-only urban percentage; LandUse is used consistently across both ...
Pass	Maharashtra-specific flag; AnalysisStatus is used consistently.
ExcludeFromSpeedSPI	Maharashtra-specific exclusion field; AnalysisStatus is used consistently.

Notebook section label: 4.3 Harmonize columns into one common schema

Standardized combined data preview (key columns)

Country	SegmentID	SourceRoad ID	OBJECTI D	Road Name	RoadClass	LandU se	SpeedLim it	MedianSpe ed	F85thPercentileSpe ed	AnalysisStat us
Maharashtra	Maharashtra 1	SE/4183	1		secondary	RURAL		0	0	Valid
Maharashtra	Maharashtra 2	SE/6376	2		secondary	RURAL	55	18	18	Valid
Maharashtra	Maharashtra 3	SE/8617	3		secondary	RURAL		23.5	27	Valid

Data-quality filtering

We keep only records that can support a defensible speed-limit review. This is important for methodological robustness because the final score should not be driven by missing or invalid inputs.

A segment is included if:

1. AnalysisStatus = Valid
2. SpeedLimit > 0
3. F85thPercentileSpeed > 0
4. RoadClass is one of: motorway, trunk, primary, secondary
5. LandUse is urban or rural

The funnel below shows how many rows remain after each validation step.

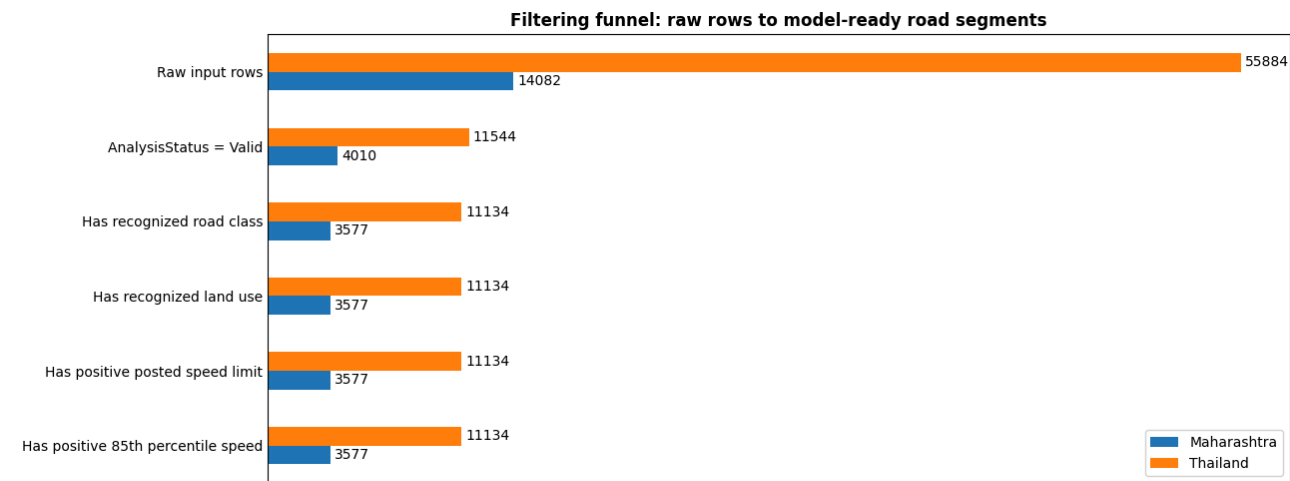
Notebook section label: 5. Filtering funnel

Filtering funnel table

Step	Maharashtra	Thailand	Total
Raw input rows	14082	55884	69966
AnalysisStatus = Valid	4010	11544	15554
Has positive posted speed limit	3577	11134	14711
Has positive 85th percentile speed	3577	11134	14711
Has recognized road class	3577	11134	14711
Has recognized land use	3577	11134	14711

Interpretation: The filtering funnel shows that most rows are removed at the validity and positive-speed checks. The final scored dataset contains 14,711 valid segments. This matters because the score should be based only on records with the minimum information needed for a defensible speed-limit review.

Notebook section label: Chart: filtering funnel by country / Calculate total to sort the y-axis (categories) descending by value / Sort ascending so that the plot (plotted from bottom to top) shows largest values at the top



Filtering funnel by country

Output interpretation: The funnel chart makes the data cleaning transparent. It shows how the dataset moves from raw rows to valid rows that can support scoring. This helps reviewers see that segments were not removed arbitrarily; they were filtered because key speed, road class, or land-use information was missing or invalid.

Notebook section label: Missingness check after filtering

Missingness after filtering

Field	Missing % after filtering
RoadClass	0
LandUse	0
SpeedLimit	0
MedianSpeed	0
F85thPercentileSpeed	0
WeightedSample	0
SampleSizeUnified	0
RankedPercentile	0
Shape_Length	0

Safe System reference speed table

The reference speed is a **screening benchmark**, not a final legal speed-limit recommendation.

The directly published Safe System anchor values commonly used in speed-management guidance are around:

- **30 km/h** where pedestrians, cyclists, motorcycles, or other vulnerable users mix with vehicles.

- **50 km/h** where side-impact / intersection risk is important.
- **70 km/h** where head-on crashes are possible and median protection is unknown or absent.
- **100 km/h** for access-controlled, protected motorway-type environments.

Because the Phase 1 dataset does not include detailed engineering features such as median barriers, school zones, crossings, roadside markets, lane width, or access control, we translate the guidance into a practical **road-class + land-use screening table**.

The model is intentionally conservative for urban and secondary roads because they are more likely to have vulnerable-road-user exposure.

Notebook section label: 6. Reference speed table

Safe System reference speed table

LandUse	RoadClass	SafeReferenceLimit_kmh	Rationale
URBAN	secondary	30	Likely pedestrian/cyclist/two-wheeler exposure and roadside activity; higher
URBAN	primary	50	Urban arterial context; side-impact and intersection risk may be present.
URBAN	trunk	50	Higher-function urban road, but still in an urban exposure environment unless access control is confirmed.
URBAN	motorway	80	Assumes stronger separation; field validation needed to confirm controlled access.
RURAL	secondary	60	Mixed traffic and roadside access may still be present.
RURAL	primary	70	Possible head-on and access risk where physical separation is unknown.
RURAL	trunk	80	Higher-function rural road; verify protection before allowing higher limits.
RURAL	motorway	100	Assumes limited access and physical protection.

Interpretation: The Safe System reference table is the policy anchor of the model. The values are not treated as final legal speed limits. They are a transparent screening reference based on land use and road class, used to ask whether the posted limit looks too high for the road context.

Notebook section label: Apply Safe System reference speeds

Safe reference mapping preview

Country	SegmentID	RoadName	RoadClass	LandUse	SpeedLimit	SafeReferenceLimit_kmh
Maharashtra	Maharashtra_2	None	secondary	RURAL	55	60
Maharashtra	Maharashtra_4	उमरी रोड	secondary	RURAL	55	60
Maharashtra	Maharashtra_5	None	secondary	RURAL	55	60
Maharashtra	Maharashtra_6	None	secondary	RURAL	45	60
Maharashtra	Maharashtra_9	None	secondary	RURAL	55	60

Feature engineering: translating the problem into measurable signals

Each segment receives five interpretable signals.

Signal	Plain-language meaning	Why it matters
`PostedExcess_kmh`	How far the posted speed limit is above the Safe System reference speed	Main measure of speed-limit misalignment.
`OperatingExcess_kmh`	Whether typical high-end operating speeds are also above the reference speed	Adds real-world context without turning the model into speeding enforcement.
`VRUExposureProxy`	Proxy for vulnerable-road-user exposure from land use and road class	Helps prioritize urban and lower-class roads where people outside cars are more exposed.
`TrafficExposureScore`	Whether the road carries more travel exposure	Helps government prioritize locations with larger potential public-safety impact.
`PeerGap_kmh`	How high the posted limit looks compared with similar peer roads	AI support signal; kept small for transparency.

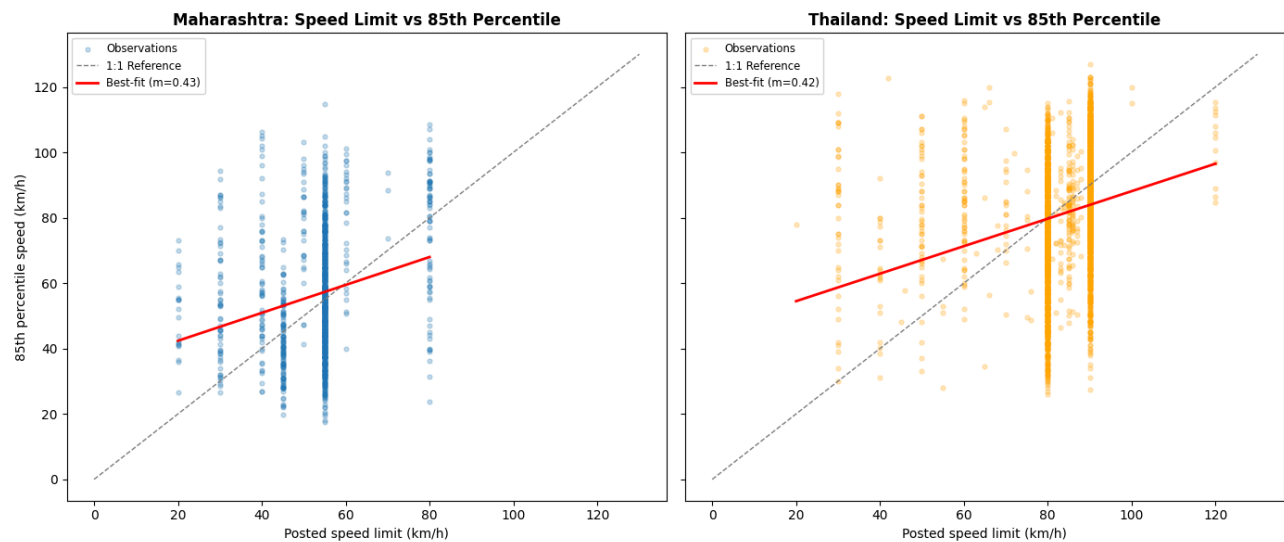
Notebook section label: 7. Feature engineering

Feature engineering preview

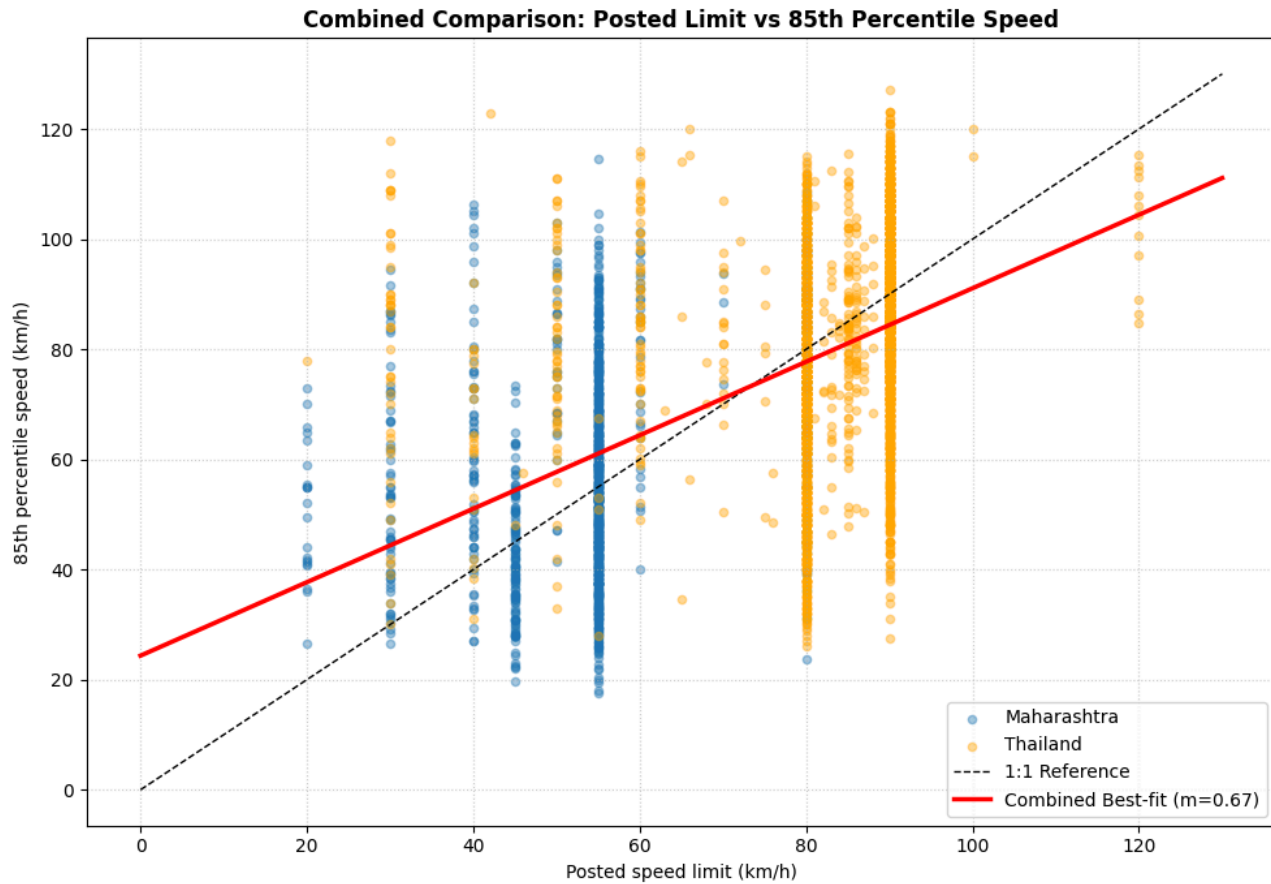
Country	Segment ID	RoadName	RoadClass	Land Use	Speed Limit	SafeReferenceLimit_kmh	PostedExcess_kmh	F85thPercentileSpeed	OperatingExcess_kmh	VRUExposureProxy	TrafficExposureScore	SampleConfidence
Maharashtra	Maharashtra_2		secondary	RURAL	55	60	0	18	0	0.45	0	0.038
Maharashtra	Maharashtra_4	उमरी रोड	secondary	RURAL	55	60	0	18	0	0.45	0	0.088
Maharashtra	Maharashtra_5		secondary	RURAL	55	60	0	13.333	0	0.45	0	0.106
Maharashtra	Maharashtra_6		secondary	RURAL	45	60	0	62.5	0	0.45	0	0.166
Maharashtra	Maharashtra_9		secondary	RURAL	55	60	0	29.5	0	0.45	0	0.193
Maharashtra	Maharashtra_10		secondary	RURAL	55	60	0	35.5	0	0.45	0	0.126
Maharashtra	Maharashtra_11		secondary	RURAL	55	60	0	43	0	0.45	0	0.12
Maharashtra	Maharashtra_13		secondary	RURAL	55	60	0	42	0	0.45	0	0.12
Maharashtra	Maharashtra_14		secondary	RURAL	55	60	0	29.333	0	0.45	0	0.178
Maharashtra	Maharashtra_15		secondary	RURAL	55	60	0	48.5	0	0.45	0	0.242

Interpretation: The preview confirms that rural secondary roads in Maharashtra often have a Safe System reference of 60 km/h. When the posted speed limit is 55 km/h, PostedExcess_kmh becomes 0 because the posted limit is not above the reference. Other signals, such as F85th percentile speed, VRU exposure, and traffic exposure, still remain available for scoring.

Notebook section label: Speed limit vs 85th percentile speed / Chart 1: Faceted Scatter with Best-Fit Lines / Explicit color: Thailand = orange, Maharashtra = blue



Posted speed limit vs 85th percentile speed



Posted speed limit vs 85th percentile speed

Chart 1: Maharashtra - Speed Limit vs 85th Percentile Speed

This chart compares the posted speed limit on Maharashtra road segments with the 85th percentile speed, which is a way of showing the speed that most drivers are staying under. In simple terms, it gives us an idea of the typical higher-end operating speed on the road.

The black dashed line is the 1:1 reference line. If a point sits exactly on this line, it means the 85th percentile speed is equal to the posted speed limit. For example, if the posted limit is 60 km/h and the 85th percentile speed is also 60 km/h, the point would fall on this line.

The blue points show the actual road observations in Maharashtra. One thing we can see is that there is a lot of spread at the same posted speed limit. For example, many roads with similar posted limits still have very different 85th percentile speeds. This tells us that the posted limit alone does not fully explain how fast drivers actually travel. Road design, roadside activity, access points, enforcement, and road condition may also affect behavior.

The red best-fit line slopes upward, which means that as posted speed limits increase, operating speeds also tend to increase. However, the slope is only around 0.43, meaning the increase is not one-to-one. So if the posted limit increases by 10 km/h, the typical high-end operating speed does not increase by the full 10 km/h. It increases, but more slowly.

A useful observation is that at lower and mid-range posted limits, some points are above the 1:1 line. This means that on some lower-speed roads, drivers are traveling faster than the posted limit. At higher posted limits, many points are closer to or below the reference line, which may suggest better compliance or that the road environment naturally limits speeds.

Chart 2: Thailand - Speed Limit vs 85th Percentile Speed

This chart shows the same comparison, but for Thailand.

The orange points are Thailand road observations. Like Maharashtra, the points are quite spread out. This means that even roads with the same posted speed limit can have very different actual operating speeds.

One very noticeable pattern is that many observations are concentrated around 80 km/h. This suggests that 80 km/h is a common posted speed limit in the Thailand dataset. But even within that same 80 km/h posted limit, the 85th percentile speeds vary widely. Some are much lower, while others are much higher.

The red best-fit line again slopes upward, with a slope of about 0.42. This is very close to Maharashtra's slope. That means Thailand also shows the same general pattern: higher posted speed limits are linked with higher operating speeds, but not in a one-to-one way.

At lower and mid-range posted limits, several points appear above the 1:1 line, which suggests possible speeding or posted limits that do not match actual driving behavior. At the higher posted limits, many points are closer to or below the reference line, though there are still some high-speed observations.

Chart 3: Combined Comparison - Maharashtra and Thailand

The third chart combines both countries in one view. The blue points represent Maharashtra, while the orange points represent Thailand.

This chart is useful because it lets us compare whether the two places show very different patterns. Overall, the two clouds overlap quite a lot. That means the relationship between posted limits and 85th percentile speeds is broadly similar in both datasets. There is no obvious pattern where one country is completely separate from the other.

The red combined best-fit line has a stronger slope, around 0.67, compared with the separate country-level charts. This means that when we look at both countries together, the overall relationship between posted limits and operating speeds becomes clearer. In general, higher posted limits are associated with higher operating speeds.

However, the red line still does not perfectly match the black 1:1 line. This means drivers do not simply follow posted limits exactly. Actual operating speed is influenced by other road conditions too.

At low to mid posted limits, many points are above the black reference line. This suggests that in lower-speed environments, actual high-end speeds can exceed the posted limit. This matters for safety because these areas may include more access points, pedestrians, cyclists, two-wheelers, or roadside activity.

At higher posted limits, especially around 80 to 120 km/h, many points are closer to or below the 1:1 line. This may mean that drivers are more aligned with posted limits in higher-speed corridors, or that the physical road environment naturally controls speed.

Key takeaways

- As posted limits increase, typical operating speeds also increase, but by a smaller margin; drivers do not fully "use" each additional km/h of posted limit, which is why the best-fit line lies below the 1:1 black reference line at higher limits.
- At low to mid limits (roughly 20-60 km/h), many points are above the 1:1 line, suggesting drivers often exceed the posted limit in these zones, while at higher limits (80-120 km/h), points tend to fall closer to or below the reference line, implying relatively better compliance or natural speed constraints.
- The overlapping clouds of blue (Maharashtra) and orange (Thailand) indicate broadly similar operating-speed behavior between the two regions, with substantial variability at each posted limit but no stark systematic difference visible in this combined view.

AI/ML peer model: a supporting signal, not the main decision rule

The peer model estimates the speed limit typically associated with comparable road segments using available context fields.

Why include this?

A purely rule-based score may miss unusual cases. The peer model adds a data-driven check: "Compared with similar roads in the same country, does this posted limit look high?"

Why keep it small?

Posted speed limits are partly shaped by local policy and engineering details not fully included in the Phase 1 files. Therefore, the peer model is used as a **5% supporting signal**, while the main score remains transparent and Safe System-aligned.

Notebook section label: 8. Country-specific Random Forest peer model / --- Tuning config (knobs) --- / Search space - trimmed for speed. max_depth=None dropped (fully-grown trees = most of the cost).

Peer-model diagnostics

Country	Rows	Tuned	CV_folds	CV_MAE	Train_MAE	Test_MAE	Train_R2	Test_R2
Thailand	11134	True	4	5.84	5.46	5.63	0.2	0.11
Maharashtra	3577	True	4	5.2	4.47	4.89	0.35	0.19

Interpretation: The peer model estimates what posted speed limit looks typical for similar road segments. Test MAE is around 5 to 6 km/h, which is useful as a supporting comparison because speed limits are often set in 5 or 10 km/h increments. The R2 values are modest, so the peer model should not dominate the final score; it is a light AI support signal rather than the main safety anchor.

Reading the peer-model diagnostics

- **MAE** means average prediction error in km/h. Lower is better.
- **R²** shows how much of posted speed-limit variation is explained by the available fields. A modest R² is expected because speed limits depend on engineering and policy details not included in the dataset.
- Because of this, the peer model is intentionally not allowed to dominate the final score.

Peer-model explanation

The peer model is a supporting model that estimates what posted speed limit is typical for a road segment based on similar road segments. In simple terms, it asks: "Compared with roads that look similar, does this road's posted speed limit look unusually high or unusually low?" This is useful because road segments can have different contexts depending on road class, land use, traffic exposure, and other available features. Instead of judging a road only against one general rule, the peer model gives an additional comparison based on patterns found in the dataset.

However, the peer model should not dominate the final safety score. Posted speed limits are not purely data-driven; they are also influenced by engineering judgment, enforcement, road design, local policy, access control, and sometimes outdated standards. Because of that, a road's "peer expected speed" is helpful as a reference, but it should not replace the Safe System reference speed. The main safety logic should still come from the Safe System signals, while the peer model acts as a secondary check.

Thailand results

For Thailand, the peer model used 11,134 rows, which means it had the larger dataset compared with Maharashtra. The model achieved a cross-validation MAE of 5.84 km/h, a training MAE of 5.46 km/h, and a test MAE of 5.63 km/h. This

means that when the model predicts the expected posted speed limit for Thailand road segments, it is usually off by around 5 to 6 km/h on unseen test data.

The training and test errors are close to each other, which is a good sign. It suggests that the model is not just memorizing the training data. However, the test R^2 is 0.11, which is quite modest. This means the model explains only about 11% of the variation in posted speed limits on the test set. In simple terms, the available features capture some pattern, but many important factors affecting posted speed limits are probably not included in the dataset.

Maharashtra results

For Maharashtra, the peer model used 3,577 rows, so it had fewer observations than Thailand. Despite having a smaller dataset, the model performed slightly better. It achieved a cross-validation MAE of 5.20 km/h, a training MAE of 4.47 km/h, and a test MAE of 4.89 km/h. This means the model's average prediction error on unseen Maharashtra data is around 5 km/h.

The model also had a higher test R^2 of 0.19, compared with Thailand's 0.11. This means the available features explain posted speed-limit patterns a little better in Maharashtra than in Thailand. There is still a gap between train R^2 of 0.35 and test R^2 of 0.19, but the gap is not alarming. It simply shows that the model performs better on data it has already seen, which is normal, while still retaining some useful predictive value on new data.

Overall results and key takeaways

Overall, the peer model performs reasonably as a supporting comparison tool. The test MAE is 5.63 km/h for Thailand and 4.89 km/h for Maharashtra, so the model is typically off by about 5 km/h. Since speed limits are often set in increments like 5 or 10 km/h, this level of error is still useful for flagging whether a road's posted limit looks higher or lower than similar roads.

The key limitation is that the R^2 values are modest: 0.11 for Thailand and 0.19 for Maharashtra. This means the model does not fully explain posted speed-limit decisions. That is expected because speed limits depend on many details not fully captured in the dataset, such as road geometry, intersections, median separation, crash history, enforcement, and local policy.

The main takeaway is that the peer model should be used as a lightweight AI support signal, not as the main decision-maker. It helps identify roads whose posted limits look unusual compared with similar roads, but the final prioritization should still rely more heavily on the Safe System features such as `PostedExcess_kmh`, `OperatingExcess_kmh`, `VRUExposureProxy`, and `TrafficExposureScore`. A good way to explain it is: "The peer model adds a data-driven comparison, but the Safe System logic remains the main safety anchor."

Final scoring model

The model converts each segment into a 0-100 `PriorityScore`. Higher means the segment should be reviewed sooner.

Score components

Component	Weight	Why this weight makes sense
Posted-limit excess	45%	The challenge is about whether the posted limit itself is appropriate.
Operating-speed excess	20%	Adds real-world speed context without blaming drivers.
VRU exposure proxy	15%	Prioritizes places where crash forces may affect pedestrians, cyclists, and two-wheelers.
Traffic exposure	15%	Gives higher priority to roads with more travel exposure.
AI peer gap	5%	Adds innovation but keeps the score explainable and policy-grounded.

Priority bands

Priority band	PriorityScore range	Meaning
Critical review	70-100	Immediate engineering review and field validation.
High priority	50-69.9	Review soon as part of corridor safety planning.
Moderate priority	30-49.9	Monitor and validate with added contextual data.
Monitor	0-29.9	Keep under normal network monitoring.

Notebook section label: 9. Final scoring / Component contributions add directly to PriorityScore.

Scoring preview

Country	SegmentID	RoadName	RoadClass	LandUse	SpeedLimit	SafeReferenceLimit kmh	F85thPercentileSpeed	PriorityScore	SpeedSafetyScore	PriorityBand	Recommendation
Maharashtra	Maharashtra_2		secondary	RURAL	55	60	18	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_4	उमरी रोड	secondary	RURAL	55	60	18	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_5		secondary	RURAL	55	60	13.333	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_6		secondary	RURAL	45	60	62.5	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_9		secondary	RURAL	55	60	29.5	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_10		secondary	RURAL	55	60	35.5	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_11		secondary	RURAL	55	60	43	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_13		secondary	RURAL	55	60	42	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_14		secondary	RURAL	55	60	29.333	6.8	93.2	Monitor	Monitor as part of normal network management
Maharashtra	Maharashtra_15		secondary	RURAL	55	60	48.5	7.2	92.8	Monitor	Monitor as part of normal network management

Results summary

Current run insight from the provided files

Using the uploaded Maharashtra and Thailand GeoJSON files, this run scores **14,711 valid road segments** covering about **99,986 km**. The validation funnel keeps **3,577 Maharashtra** segments and **11,134 Thailand** segments after requiring valid analysis status, positive posted speed limit, positive F85 speed, recognized road class, and recognized land use.

Country	Valid segments	Length km	Critical review	High priority	Moderate priority	Monitor	Key interpretation
Maharashtra	3,577	40,266	17	83	128	3,349	Most segments are monitor-level because posted limits are generally closer to the screening reference speeds in the provided data.

Thailand	11,134	59,721	6,080	2,503	1,670	881	The model finds a much larger concentration of review needs, especially on urban and mixed-use roads with posted limits above the urban reference speeds.
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Overall, **8,683 segments** fall into Critical or High priority, covering about **45,363 km**. This is the recommended shortlist for engineering review and field validation, not an automatic legal speed-limit change list.

Notebook section label: 10. Results summary tables

Summary by country

Country	Segments	Length km	MedianPriorityScore	MeanPriorityScore	MedianSpeedLimit	MedianF85
Maharashtra	3577	40266	6.8	10.2	55	54
Thailand	11134	59721	73.2	65.4	88	83

Interpretation: Thailand has more valid segments and a higher median priority score than Maharashtra. The Thailand dataset also has higher median posted and operating speeds, which helps explain why more segments move into higher review categories under the Safe System-based scoring logic.

Priority-band counts by country

PriorityBand	Country	Critical review	High priority	Moderate priority	Monitor
0	Maharashtra	17	83	128	3349
1	Thailand	6082	2497	1673	882

Interpretation: Maharashtra is mostly in the Monitor band, while Thailand has many more Critical review and High priority segments. This should be read as a prioritization result based on the available features, not as a final judgment that one road network is safe or unsafe.

Priority-band length by country

PriorityBand	Country	Critical review	High priority	Moderate priority	Monitor
0	Maharashtra	56	211	234	39764
1	Thailand	28100	16982	10098	4541

Interpretation: The length table checks whether high-priority findings are just many short segments or a meaningful length of road. Thailand has a large amount of road length in Critical review and High priority, which suggests a broader review scope. Maharashtra has most of its length in Monitor.

Country summary results

The first table gives the overall profile of the road segments included in the scoring. Maharashtra has 3,577 segments, covering around 40,266 km, while Thailand has 11,134 segments, covering around 59,721 km. This means Thailand has a much larger sample in the dataset, both in number of segments and total road length.

The priority scores are very different between the two countries. Maharashtra has a median priority score of 6.8 and a mean priority score of 10.2, which suggests that most Maharashtra segments are relatively low priority, with only a few

higher-scoring segments pulling the average upward. Thailand has a much higher median priority score of 73.2 and mean priority score of 65.4, meaning a large share of Thailand's segments are being scored as higher concern for review.

The speed values help explain part of this difference. Maharashtra's median posted speed limit is 55 km/h, and its median 85th percentile speed is 54 km/h, so typical operating speed is close to the posted limit. Thailand's median posted speed limit is 88 km/h, and its median 85th percentile speed is 83 km/h, which shows that the Thailand dataset contains many higher-speed road segments. Because higher speeds increase safety concern under the Safe System logic, this likely contributes to Thailand's higher priority scores.

Priority count results

The second table counts how many road segments fall into each priority band: Critical review, High priority, Moderate priority, and Monitor.

For Maharashtra, most segments fall under Monitor. Out of 3,577 segments, 3,349 are only for monitoring, while only 17 are Critical review, 83 are High priority, and 128 are Moderate priority. This means the scoring model is not flagging Maharashtra broadly as high concern. Instead, it identifies only a small number of segments that may need closer safety review.

For Thailand, the pattern is almost the opposite. Out of 11,134 segments, 6,082 are classified as Critical review, 2,497 as High priority, 1,673 as Moderate priority, and only 882 as Monitor. This means the model flags a large portion of Thailand's segments for more urgent review. This may be due to higher posted speed limits, higher operating speeds, road-class composition, or how the Safe System reference speed compares with the observed road context.

Priority length results

The third table is similar to the second table, but instead of counting the number of segments, it sums the total kilometers in each priority band. This is important because one segment can be short while another can cover a much longer road length.

For Maharashtra, the length results confirm the same pattern as the segment counts. Around 39,764 km are in the Monitor category, while only 56 km are in Critical review, 211 km are in High priority, and 234 km are in Moderate priority. This means that not only are most Maharashtra segments low priority by count, but most of the total road length is also low priority.

For Thailand, the length results again show a heavier concentration in the higher-priority categories. About 28,100 km are in Critical review, 16,982 km are in High priority, 10,098 km are in Moderate priority, and only 4,541 km are in Monitor. This means the model is not just flagging many short segments; a large amount of total road length is also being prioritized for review.

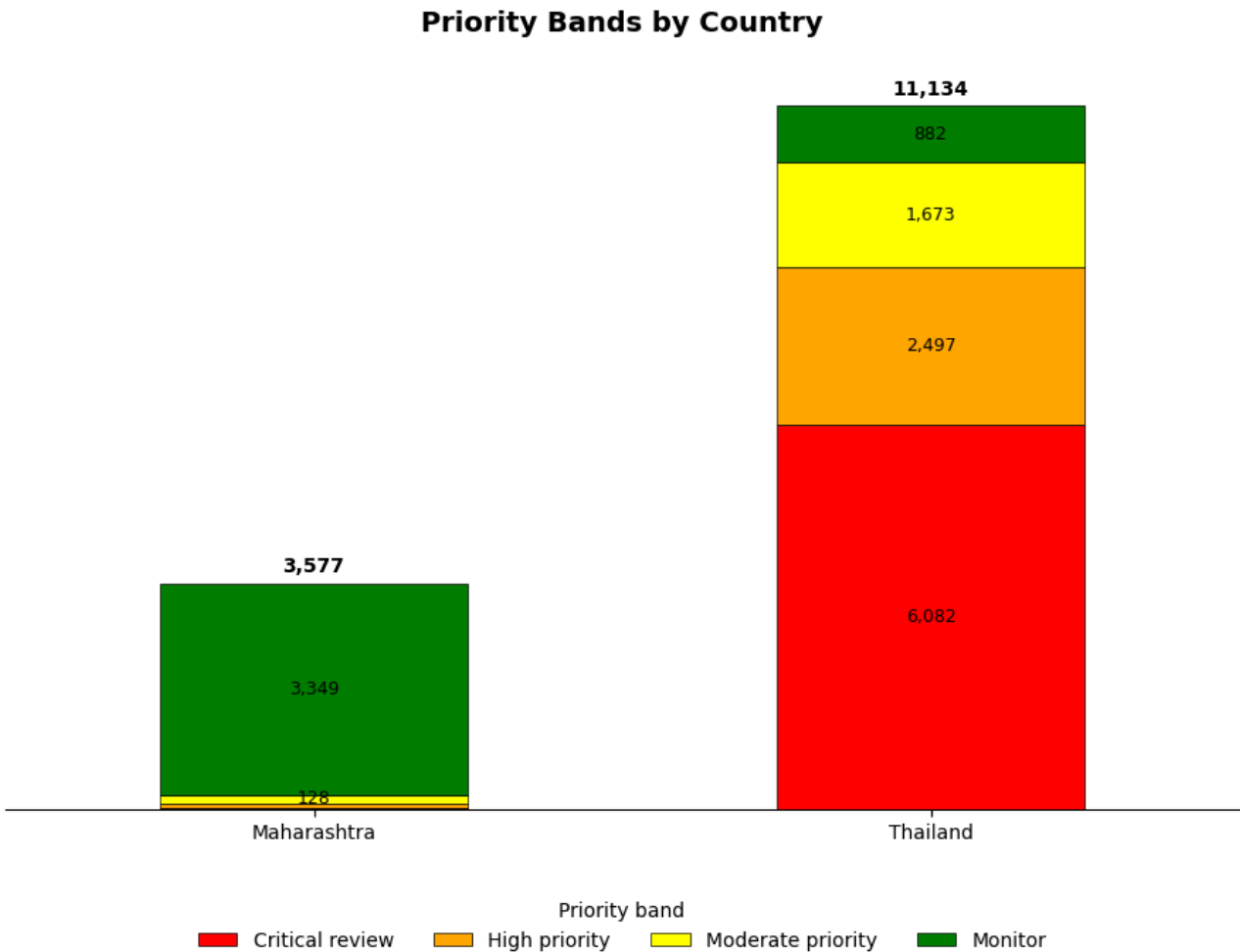
This matters because road safety prioritization should consider both how many segments are flagged and how much road length those segments represent. A country may have many flagged segments, but if they are very short, the review burden is different. In Thailand's case, both the segment count and total kilometers suggest a large review scope.

Overall results and key takeaways

Overall, the results show a clear contrast between Maharashtra and Thailand. Maharashtra is mostly in the Monitor category, with only a small number of segments and kilometers requiring critical or high-priority review. Its median speed limit and median 85th percentile speed are also close to each other, which suggests fewer extreme mismatches between posted limits and typical operating speeds in the displayed summary.

Thailand, on the other hand, has much higher priority scores and a much larger share of segments in the Critical review and High priority categories. Its median posted speed limit and median 85th percentile speed are also much higher than Maharashtra's, which likely contributes to the higher priority scores under the Safe System-based scoring logic.

Notebook section label: Chart: priority bands by country / Custom color mapping as requested / Set title and remove labels per request

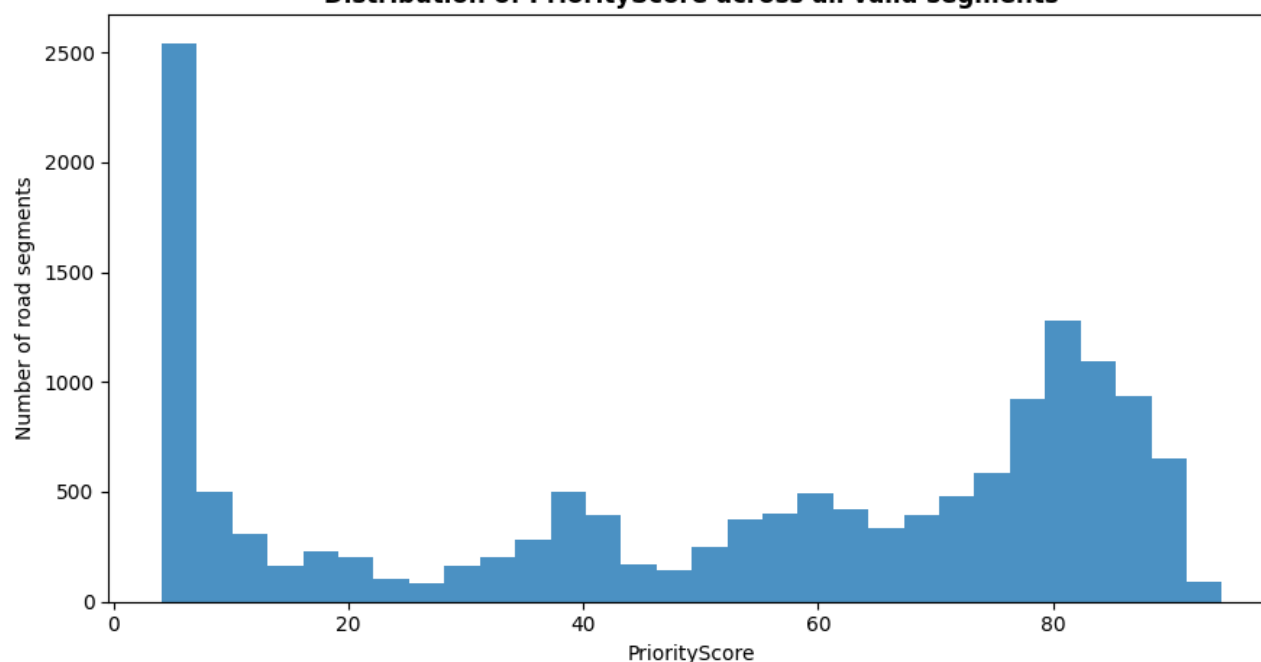


Priority bands by country

Output interpretation: The priority band chart turns the score into a practical review queue. It makes the country-level difference visible: Maharashtra is dominated by Monitor segments, while Thailand has a much larger share of Critical review and High priority segments.

Notebook section label: Chart: priority score distribution

Distribution of PriorityScore across all valid segments



Distribution of PriorityScore across valid road segments

Output interpretation: The PriorityScore histogram shows a two-sided pattern. Many segments fall into the low-score range, while another large group falls into the high-score range. This means the model is not assigning every road the same priority; it is separating routine monitoring cases from roads that deserve closer review.

The PriorityScore distribution shows a two-sided pattern. A large number of segments fall in the low-score range, indicating roads that mainly require routine monitoring. At the same time, there is a clear concentration of high-scoring segments around the upper end of the distribution, suggesting that the model identifies a substantial group of roads for more urgent review. This supports the usefulness of the scoring framework because it separates segments into different priority levels rather than assigning similar scores to all roads. The results should be interpreted as a prioritization tool for review, not as a final safety judgment.

Notebook section label: Top combinations by priority band: useful to understand where the strongest pattern comes from.

Top combinations by priority band

Country	LandUse	RoadClass	PriorityBand	Segments	Length_km	MedianPriority
Maharashtra	RURAL	secondary	Monitor	1190	11198	6.9
Maharashtra	RURAL	trunk	Monitor	877	15697	4.4
Maharashtra	RURAL	primary	Monitor	743	11626	5.9
Maharashtra	URBAN	primary	Monitor	305	614	11.1
Maharashtra	URBAN	trunk	Monitor	197	561	9.6
Maharashtra	URBAN	secondary	Moderate priority	120	196	38.8
Maharashtra	URBAN	secondary	High priority	46	97	63
Maharashtra	URBAN	motorway	Monitor	23	51	8.4
Maharashtra	URBAN	trunk	High priority	21	66	59.6
Maharashtra	URBAN	primary	High priority	16	47	63.6
Maharashtra	URBAN	trunk	Critical review	15	50	76.9
Maharashtra	URBAN	secondary	Monitor	10	12	12.1
Maharashtra	URBAN	trunk	Moderate priority	6	32	32
Maharashtra	RURAL	motorway	Monitor	4	6	6.4
Maharashtra	RURAL	primary	Moderate priority	2	6	31.6
Maharashtra	URBAN	primary	Critical review	2	6	72.8
Thailand	URBAN	secondary	Critical review	2440	5805	81.9
Thailand	URBAN	primary	Critical review	1441	3795	82.7
Thailand	RURAL	secondary	Critical review	1302	13066	76.8
Thailand	RURAL	secondary	High priority	1209	10731	58.9

Top-priority segments and why they were flagged

This table is designed to be used directly in a submission or policy discussion. It shows not just which roads rank highest, but **why** they rank highly.

Notebook section label: 11. Top segments with explainable component contributions

Top-priority road segments - identification and speed context

Country	Segment ID	RoadName	RoadClass	LandUse	Speed Limit	SafeReference Limit_kmh	F85thPercentileSpeed	PostedExcess_kmh	Operating Excess_kmh	PriorityScore	PriorityBand	Recommendation
Thailand	Thailand_104	Kanchanaphisek Road	trunk	URBAN	120.0	50	86.4	70	36.4	94.4	Critical review	Immediate engineering review and field validation
Thailand	Thailand_34041	Phraek Sa Road	secondary	URBAN	90	30	65.5	60	35.5	94.1	Critical review	Immediate engineering review and field validation
Thailand	Thailand_35526	7 Frontage Road	secondary	URBAN	90	30	73	60	43	94	Critical review	Immediate engineering review and field validation
Thailand	Thailand_35527	7 Frontage Road	secondary	URBAN	90	30	73.5	60	43.5	93.9	Critical review	Immediate engineering review and field validation
Thailand	Thailand_34948	Phahon Yothin Road	secondary	URBAN	90	30	72.5	60	42.5	93.9	Critical review	Immediate engineering review and field validation
Thailand	Thailand_35440	Kanchanaphisek Road Outer Service Road	secondary	URBAN	90	30	69.5	60	39.5	93.7	Critical review	Immediate engineering review and field validation
Thailand	Thailand_15711	Borommaratchonnani Road (frontage)	secondary	URBAN	90	30	74.5	60	44.5	93.7	Critical review	Immediate engineering review and field validation
Thailand	Thailand_405	Chalermprakit Ror 9	secondary	URBAN	90	30	76	60	46	93.6	Critical review	Immediate engineering review and field validation
Thailand	Thailand_55762	Bang Kruai-Sai Noi Road	secondary	URBAN	90	30	61.5	60	31.5	93.6	Critical review	Immediate engineering review and field validation
Thailand	Thailand_35435	Kanchanaphisek Road Outer Service Road	secondary	URBAN	90	30	83.8	60	53.8	93.6	Critical review	Immediate engineering review and field validation
Thailand	Thailand_55871	Rama II Road	secondary	URBAN	90	30	75	60	45	93.6	Critical review	Immediate engineering review and field validation
Thailand	Thailand_49220		secondary	URBAN	90	30	79.5	60	49.5	93.4	Critical review	Immediate engineering review and field validation
Thailand	Thailand_55759	Bang Kruai-Sai Noi Road	secondary	URBAN	90	30	69	60	39	93.3	Critical review	Immediate engineering review and field validation
Thailand	Thailand_35880	Kanchanaphisek Road Inner Service Road	secondary	URBAN	90	30	62	60	32	93.3	Critical review	Immediate engineering review and field validation
Thailand	Thailand_34175	Bangkok-Chonburi Motorway	motorway	URBAN	120.0	80	111.5	40	31.5	93.2	Critical review	Immediate engineering review and field validation

Table excerpt only: first 15 of 25 rows are shown. The full table is exported in the notebook output file.

Top-priority road segments - contribution breakdown

SegmentID	PostedExcessContribution	OperatingExcessContribution	VRUContribution	TrafficContribution	PeerGapContribution	SpeedSafetyScore	StreetImageLink
Thailand_104	45	20	9.4	15	5	5.6	100.4107064,13.6417155,100.43193453,13.96699621
Thailand_34041	45	20	12	14.5	2.6	5.9	100.6088384,13.5870537,100.6931267,13.561889
Thailand_35526	45	20	12	14.4	2.6	6	100.9798512,13.1174787,100.9901822,13.1695521
Thailand_35527	45	20	12	14.2	2.6	6.1	100.9936166,13.1702103,100.9854164,13.1266561
Thailand_34948	45	20	12	14.7	2.2	6.1	100.6414464,14.1924122,100.76665496,14.25554596
Thailand_35440	45	20	12	14	2.7	6.3	100.69614827,14.07532886,100.7080897,14.0089861
Thailand_15711	45	20	12	14.2	2.4	6.3	100.2472605,13.7835041,100.3248097,13.7877394
Thailand_405	45	20	12	13.6	3	6.4	98.38654376,7.95027207,98.3752149,7.9320134
Thailand_55762	45	20	12	14.5	2.1	6.4	100.32498932,13.98124145,100.4106348,13.9269943
Thailand_35435	45	20	12	14.6	2.1	6.4	100.6912026,13.896726,100.6752938,13.8554855
Thailand_55871	45	20	12	13.7	2.8	6.4	100.1813193,13.5237393,100.1888763,13.5262981
Thailand_49220	45	20	12	13.8	2.7	6.6	100.9901822,13.1695521,100.9936166,13.1702103
Thailand_55759	45	20	12	14.1	2.3	6.7	100.4105249,13.9269181,100.32498932,13.98110235
Thailand_35880	45	20	12	14	2.4	6.7	100.7020097,14.0064776,100.6954063,14.07532886
Thailand_34175	45	20	8.2	15	5	6.8	100.93959082,13.60866263,101.0021552,13.3077079

Table excerpt only: first 15 of 25 rows are shown. The full table is exported in the notebook output file.

Interpretation: The top-priority list is designed for policy and engineering review. The highest-ranked roads usually combine large posted-limit excess, high operating-speed context, urban or vulnerable-road-user exposure, and meaningful travel exposure. This is more useful than simply listing the highest posted speed limits because it explains why each road is flagged.

Interpretation of the top-priority list

A segment is usually high priority when several things happen together:

1. The posted limit is much higher than the Safe System reference speed.
2. The road is urban or otherwise likely to have vulnerable-road-user exposure.
3. The 85th percentile speed is also high relative to the reference.
4. The road has meaningful travel exposure.

This makes the output more useful than a simple "highest speed limit" list because it links speed limits to **road context and public-safety relevance**.

Sensitivity check: are the rankings stable?

A winning submission should show that the method is not overly fragile. Here we test two alternative weighting scenarios:

1. **Safety-heavy scenario:** gives more weight to posted-limit mismatch and VRU exposure.
2. **Exposure-heavy scenario:** gives more weight to traffic exposure.

We compare each scenario with the baseline using rank correlation and overlap in the top 300 priority segments.

Notebook section label: 12. Sensitivity analysis

Sensitivity analysis results

Scenario	Spearman rank correlation vs baseline	Top 300 overlap count	Top 300 overlap %
Safety-heavy	0.985	191	63.7
Exposure-heavy	0.998	269	89.7

Interpretation: The sensitivity test shows that the ranking is stable overall. Spearman correlations remain very high. The top 300 list changes more in the Safety-heavy scenario because that version emphasizes speed mismatch and vulnerable-road-user exposure more strongly. The Exposure-heavy scenario stays closer to the baseline, meaning the baseline already accounts for exposure in a way that is consistent with the alternative weighting.

What this sensitivity analysis does

This section checks whether the final PriorityScore ranking is stable when the weights are slightly changed. In simple terms, it asks:

"If we change the importance of the scoring components, do we still mostly flag the same road segments?"

This is important because the PriorityScore is built from several signals: posted speed excess, operating speed excess, vulnerable road user exposure, traffic exposure, and peer gap. Since the weights are partly based on project judgment, we need to test whether the results are too sensitive to those choices.

The baseline weighting is:

Signal Baseline weight

Posted speed excess 45%

Operating speed excess 20%

VRU exposure 15%

Traffic exposure 15%

Peer gap 5%

So the baseline score gives the most importance to posted speed-limit misalignment, then operating speed, then exposure signals, with peer comparison kept small.

What the two scenarios mean

The code tests two alternative scenarios:

Safety-heavy scenario

Signal Weight

Posted speed excess 50%

Operating speed excess 20%

VRU exposure 20%

Traffic exposure 5%

Peer gap 5%

This version puts more emphasis on Safe System safety logic, especially posted speed excess and vulnerable road user exposure. It reduces the traffic exposure weight. In simple terms, this scenario asks:

"What if we care more about speed mismatch and vulnerable road users than road exposure?"

Exposure-heavy scenario

Signal Weight

Posted speed excess 40%

Operating speed excess 20%

VRU exposure 15%

Traffic exposure 20%

Peer gap 5%

This version gives more weight to traffic exposure, meaning roads with greater travel exposure become more important. In simple terms, this scenario asks:

"What if we prioritize roads where more travel or road-user exposure occurs?"

How the comparison is measured

The code compares each scenario against the baseline in two ways.

First, it uses Spearman rank correlation. This checks whether the overall ranking of road segments stays similar. A value close to 1.0 means the ranking is almost the same as the baseline. A lower value would mean the ranking changed a lot.

Second, it checks the Top 300 overlap. This looks at the 300 highest-priority road segments under the baseline and asks:

"How many of those same 300 road segments are still in the top 300 under the new scenario?"

This is very practical because decision-makers will usually focus on the top priority list, not every single road in the dataset.

Safety-heavy result

The Safety-heavy scenario has a Spearman rank correlation of 0.985 versus the baseline. That is very high, which means the overall ranking remains almost the same even after increasing the weight of posted speed excess and VRU exposure.

However, the Top 300 overlap is 191 out of 300, or 63.7%. This means that around two-thirds of the baseline's top 300 road segments are still selected, but about one-third changed.

This makes sense because the Safety-heavy version reduces traffic exposure and gives more importance to road safety mismatch and vulnerable road users. So some roads that were high-priority mainly because of exposure may move down, while roads with stronger Safe System concerns may move up.

Exposure-heavy result

The Exposure-heavy scenario has a Spearman rank correlation of 0.998 versus the baseline. This is extremely close to 1, so the ranking is almost identical to the baseline.

The Top 300 overlap is 269 out of 300, or 89.7%. This means almost 90% of the top 300 road segments under the baseline are still in the top 300 under the Exposure-heavy scenario.

This suggests that increasing the traffic exposure weight slightly does not drastically change the priority list. The baseline already included traffic exposure at 15%, so raising it to 20% keeps the results mostly consistent.

Overall interpretation and key takeaway

Overall, the sensitivity analysis shows that the scoring system is reasonably robust. The rankings do not collapse or completely change when the weights are adjusted. Both scenarios have very high Spearman correlations, meaning the overall road-segment ordering remains mostly consistent.

The main difference is in the Top 300 list. The Exposure-heavy scenario keeps almost the same top review list as the baseline, with 89.7% overlap. The Safety-heavy scenario changes the top list more, with 63.7% overlap, because it shifts priority toward roads with stronger Safe System and vulnerable-user concerns.

Geospatial outputs and interactive map

The challenge asks for spatial/map-based outputs. This section exports:

1. Full scored CSV
2. Full scored GeoJSON
3. Top 300 priority CSV
4. Summary by country
5. Priority crosstab
6. Peer-model diagnostics
7. Interactive HTML map of the top priority road segments

The map is saved as an HTML file instead of embedded directly in the notebook to keep the notebook lightweight and easy to submit.

Notebook section label: 13. Save tabular and geospatial outputs / Carefully choose submission columns so outputs are useful but not unnecessarily cluttered. / Ensure all output columns exist.

Output file guide

Output file	Purpose
speed_safety_scored_segments.csv	Full scored model output for every valid road segment
top_300_priority_segments.geojson	GIS-ready top-priority road layer for immediate map review
top_300_priority_segments.csv	Immediate review list for policy and engineering teams
model_summary_by_country.csv	Country-level summary
priority_crosstab_counts.csv	Priority-band counts by country
priority_crosstab_length_km.csv	Priority-band length by country
peer_model_metrics.csv	AI peer-model diagnostics
filter_funnel.csv	Data validation and filtering audit
sensitivity_check.csv	Weight sensitivity and ranking stability check

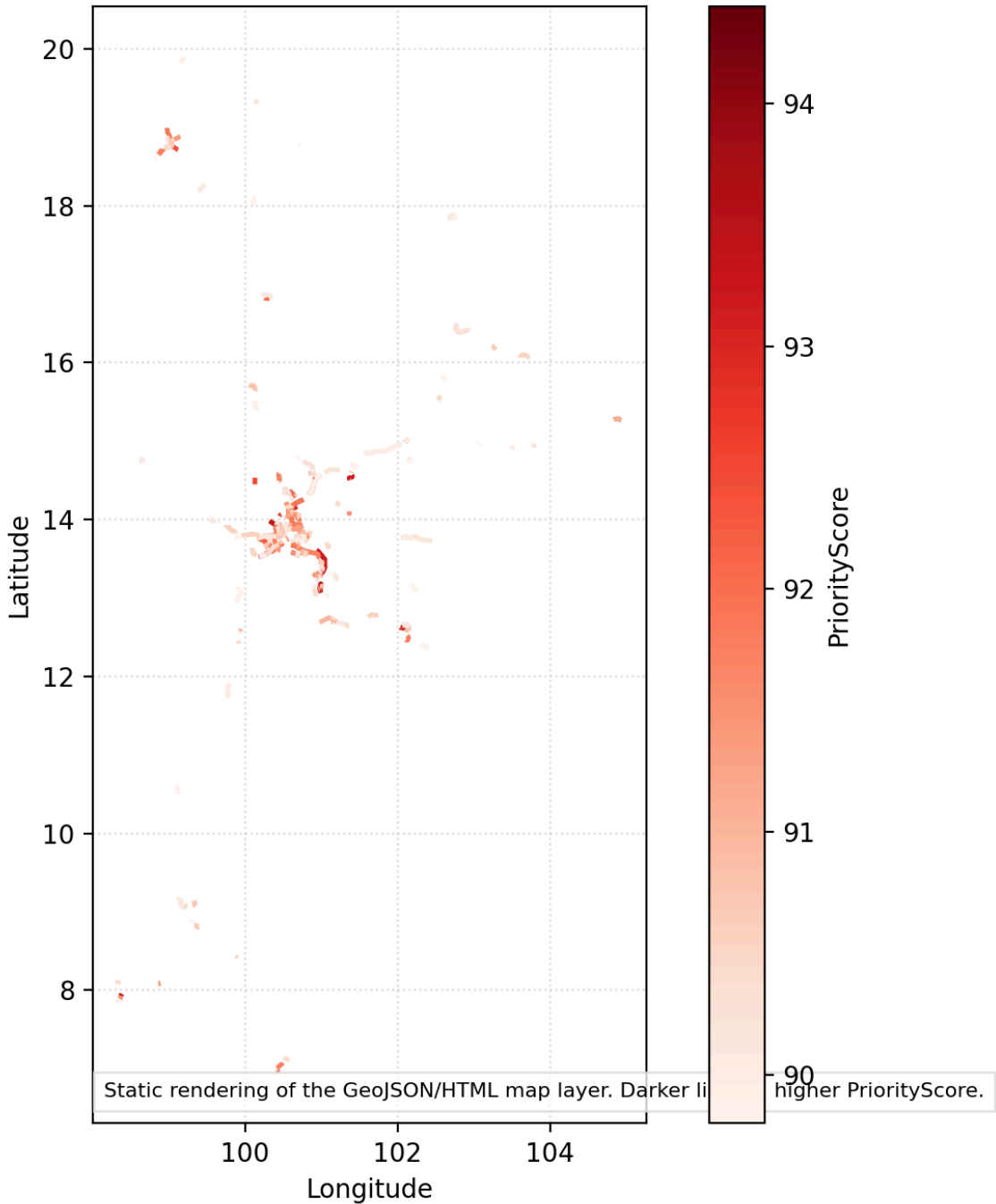
Interpretation: The output file guide shows that the notebook creates both analytical outputs and map-ready outputs. The CSV files support audit and policy review, while the GeoJSON and HTML map support geospatial exploration and immediate road-segment review.

Notebook section label: 13.2 Create an interactive map of all segments with filters / Prepare the full dataset for mapping / Ensure CRS is EPSG:4326 for Folium

Map export message

Interactive map with all segments saved to:
/content/safer_roads_submission_outputs/full_network_priority_map.html

Map view: Top 300 priority road segments (Thailand)



Map interpretation: The map-view output is the bridge between the model and field review. It lets a transport ministry or engineering team locate the highest-priority segments, filter by priority band, and inspect the reason each road was flagged. This is important because a speed-safety score is more actionable when reviewers can see exactly where the segment is and compare it with nearby road context.

Web-Based Decision Support Dashboard

To make the model outputs easier to understand and use beyond the notebook, we also developed a web-based decision support dashboard titled AI for Safer Roads – Speed Limit Misalignment Dashboard. The dashboard translates the technical scoring outputs into an interactive review environment where users can explore road segments spatially, compare model outputs, and test policy assumptions before moving to field validation. This is important because the challenge is not only about building a model, but also about making the results understandable and usable for government agencies and transport planners. ADB's challenge specifically asks for tools that assess Safe System alignment, identify risky or misaligned road segments, and produce map-based outputs that can guide review or intervention.

The site works as a practical bridge between the data science model and real-world road safety decision-making. Instead of showing only a static score or table, the dashboard allows users to inspect road segments through a map-based interface, filter results by country, road class, land use, priority band, and other segment attributes, and view the reason why a road was flagged. This makes the output more transparent because users can see not only which roads are high priority, but also whether the concern comes from the posted speed limit, Safe System reference speed, operating speed context, traffic exposure, or vulnerable-road-user exposure proxy. This supports the challenge's emphasis on policy-ready outputs and clear geospatial communication.

The dashboard also expands the modelling workflow by allowing model and scenario comparison. While the submitted methodology uses a country-specific Random Forest model only as a light supplemental peer-model layer, the site is designed to show that other modelling approaches may also be explored. This prevents the Random Forest from being presented as the sole decision-maker. Instead, the dashboard positions machine learning as one part of a wider decision-support system, where model outputs can be compared against the more transparent Safe System-based scoring logic. This is consistent with our approach of keeping the AI component supportive rather than dominant, since legal speed-limit decisions still require engineering judgment, field checks, road design information, and local policy context. The existing methodology already explains that the peer model is used as a small supporting signal, while the main score remains Safe System-aligned and interpretable.

Another important feature is the what-if analysis capability. This allows users to test how the priority ranking changes when policy assumptions or score weights are adjusted. For example, a reviewer can examine whether the priority list changes when more weight is given to Safe System mismatch, traffic exposure, or vulnerable-road-user risk. This is useful because different agencies may have different immediate goals: some may want to focus first on the most severe speed-limit mismatches, while others may prioritize corridors where more people are exposed. The what-if feature therefore makes the model more flexible without hiding the assumptions behind a black-box result. It also strengthens the robustness of the submission because the model can show whether top-priority roads remain stable under alternative policy scenarios. The notebook already includes a sensitivity check and ranking-stability output, and the dashboard makes this type of analysis easier for non-technical users to explore.

Overall, the website improves the submission by turning the notebook outputs into an operational review tool. The notebook proves that the method is reproducible from raw GeoJSON files to scored outputs, while the dashboard shows how those outputs can be used by actual decision-makers. Together, they support the major evaluation criteria of the challenge: methodological soundness, scalability, interpretability, policy relevance, and visualization clarity. The dashboard should therefore be presented not as a replacement for engineering review, but as a first-screening and prioritization tool that helps agencies decide where to look first, what issue to investigate, and which road segments may need closer field validation

Key insights for the submission narrative

1. **The model solves the correct problem.** It evaluates whether posted speed limits appear aligned with Safe System principles and road context, rather than identifying individual speeding behavior.
2. **The results are actionable.** The output is not only a score; it gives a ranked list, priority band, recommendation, map layer, and interactive map.
3. **The method is scalable.** The same standardized field schema works across Maharashtra and Thailand, despite different original field names.
4. **The model is transparent.** Every road segment can be explained through component contributions.
5. **The AI/ML is used responsibly.** The peer model adds innovation but is weighted lightly because legal speed limits require engineering details beyond the available dataset.
6. **The model is designed for refinement.** In the next phase, it can absorb street-level imagery, school/market proximity, crash history, intersection density, medians/barriers, and pedestrian crossing data.

Notebook section label: Create a concise executive summary from the model outputs.

Critical/high executive summary by country

Country	Critical High Segments	Critical High Length km	Median Priority
Maharashtra	100	267	64.6
Thailand	8579	45082	78.8

Executive summary printed by the notebook

```
Executive summary
- Valid scored segments: 14711
- Total scored length (km): 99986.0
- Critical/high segments: 8679
- Critical/high length (km): 45349.0
```

Critical/high summary by country:

Recommended additions before final challenge submission

Highest-impact additions

1. **Add Mapillary / street-level imagery features** if the challenge dataset provides usable image features or URLs. Examples: pedestrian crossing, median barrier, lane count, shoulder, roadside activity, signs.
2. **Add school, market, hospital, transit stop, and population-density layers** to improve the vulnerable-road-user exposure proxy.
3. **Add crash history** if available, especially fatal and serious injury crashes, to validate whether flagged roads align with observed harm.

4. **Add intersection density and access-control indicators** to better distinguish urban arterials from protected motorways.
5. **Add a reviewer-facing dashboard or map story** showing top segments, why they were flagged, and what intervention type may be appropriate.

Potential intervention tags for the next version

The model can later recommend possible intervention categories such as:

- speed-limit review,
- traffic calming,
- pedestrian crossing review,
- median/barrier assessment,
- school/market-zone treatment,
- enforcement support only after engineering review.

Final submission positioning

We developed a reproducible AI/ML and geospatial screening model that identifies road segments where posted speed limits may be misaligned with Safe System principles, road function, land use, observed operating speeds, vulnerable-road-user exposure, and travel exposure. The model produces a Speed Safety Score, Priority Score, priority band, recommendation, GIS layer, and interactive map. It is designed to help governments prioritize engineering review and field validation, not to automatically change legal speed limits or blame individual drivers.

The web-based dashboard strengthens the final submission by packaging the model outputs into a practical decision-support interface, where users can explore priority segments on a map, compare modelling assumptions, run what-if scenarios, and understand why each road was flagged before moving to engineering validation.

The main strength of the approach is that it is **transparent enough for policy users**, **reproducible enough for technical reviewers**, and **scalable enough to apply across countries** with similar mobility and road-network datasets.

Notebook section label: 16. Final check: list all created files

Final created files list

Created file	Size MB
01_filtering_funnel.png	0.076
02a_speedlimit_vs_f85_by_country.png	0.212
02b_speedlimit_vs_f85_combined.png	0.251
03_priority_bands_by_country.png	0.041
04_priority_score_distribution.png	0.041
filter_funnel.csv	0
full_network_priority_map.html	83.489
model_summary_by_country.csv	0
peer_model_metrics.csv	0
priority_crosstab_counts.csv	0
priority_crosstab_length_km.csv	0
sensitivity_check.csv	0
speed_safety_scored_segments.csv	4.93
top_300_priority_segments.csv	0.105
top_300_priority_segments.geojson	1.842

Appendix A. Notebook code-cell index

This appendix preserves the code-cell structure of the notebook without repeating every line of code. The full executable code remains in the submitted .ipynb file. The purpose here is to show that every implementation block has a traceable role in the methodology.

Notebook code-cell index

Cell	Notebook code section	Output type
4	2.1 Setup	printed output
5	2.2 Locate input files	printed output
7	3. Load data	table/map
8	Quick column audit: which fields appear in each file?	table/map
10	4.1 Included / standardized fields dictionary	table/map
11	4.2 Excluded fields dictionary	table/map
12	4.3 Harmonize columns into one common schema	printed output, table/map
14	5. Filtering funnel	table/map
15	Chart: filtering funnel by country	chart
16	Missingness check after filtering	table/map
18	6. Reference speed table	table/map
19	Apply Safe System reference speeds	table/map
21	7. Feature engineering	table/map
22	Speed limit vs 85th percentile speed	chart
26	8. Country-specific Random Forest peer model	printed output, table/map
30	9. Final scoring	table/map
33	10. Results summary tables	table/map
34	Code cell	table/map
35	Code cell	table/map
37	Chart: priority bands by country	chart
38	Chart: priority score distribution	chart
40	Top combinations by priority band: useful to understand where the strongest pattern comes from.	table/map
42	11. Top segments with explainable component contributions	table/map
45	12. Sensitivity analysis	table/map
48	13. Save tabular and geospatial outputs	table/map
49	13.2 Create an interactive map of all segments with filters	printed output, table/map
51	Create a concise executive summary from the model outputs.	printed output, table/map
54	16. Final check: list all created files	table/map

Appendix B. Exported output package guide

Exported output file guide

Created file	Size MB	Use in submission
01_filtering_funnel.png	0.076	Chart showing data retention through the filtering funnel.
02a_speedlimit_vs_f85_by_country.png	0.212	Country charts comparing posted limits and 85th percentile speeds.
02b_speedlimit_vs_f85_combined.png	0.251	Combined comparison of posted limits and operating speeds.
03_priority_bands_by_country.png	0.041	Country comparison of priority bands.
04_priority_score_distribution.png	0.041	Histogram of PriorityScore across valid segments.
filter_funnel.csv	0	Data validation and filtering audit.
full_network_priority_map.html	83.529	Interactive map of scored road segments with filters and popups.
model_summary_by_country.csv	0	Country-level summary of segment counts, road length, speeds, and priority scores.
peer_model_metrics.csv	0	AI peer-model diagnostics.
priority_crosstab_counts.csv	0	Priority-band counts by country.
priority_crosstab_length_km.csv	0	Priority-band length totals by country.
sensitivity_check.csv	0	Weight sensitivity and ranking stability check.
speed_safety_scored_segments.csv	4.93	Full scored model output for every valid road segment.
top_300_priority_segments.csv	0.106	Immediate review list for policy and engineering teams.
top_300_priority_segments.geojson	1.881	GIS-ready top-priority road layer for map review.

References and source notes

- ADB AI for Safer Roads Innovation Challenge official page and selection criteria.
- AI for Safer Roads 2026 Data User Guide v1.0.
- WHO, World Bank / GRSF, ITF/OECD, and iRAP Safe System and speed-management guidance used as narrative anchors for Safe System speed reasoning.
- Uploaded notebook: 260625_Feature_Engineers.ipynb.
- Exported output package: Outputs-20260701T104159Z-3-001.zip.